

phones operating on the European GSM standard (the phone number is the prefix of the e-mail address in the message), the Trojan tried to spam people with SMS messages—in this case a derogatory depiction of Spanish telecom provider Telefonica Moviles.

Like the Liberty Crack Trojan, the Timofonica attack was benign and caused little damage. Although the program reached out into the wireless world, it propagated via land-based PCs and e-mails, not from phone to phone directly. Nevertheless, Timofonica demonstrated in-the-wild, the ability of malicious code to tap into the wireless infrastructure and spread with great speed. Timofonica had the potential to flood the wireless network with messages, reducing its performance or even impairing its ability to meet load. Worse, for wireless users billed on a per-message basis, receiving spam costs them money. A similar program was observed on Japan's ambitious I-mode system. Japan's largest cellular phone maker, NTT DoCoMo, developed and owns the I-mode system which appears to have successfully captured both consumer and business markets for wireless device transactions, wireless Internet access, and instant messaging in Japan. With more than 10 million users only 18 months after its launch, some analysts see I-mode as a feasible alternative to WAP being used in Europe and touted in North America.

In June 2000, a piece of malicious code began to send a particular message to wireless users on the I-mode system. When the user received the message and clicked on a hypertext link, the program dialled 110—the Japanese equivalent of 911 in North America—without the prior knowledge of the user. This loading of emergency service lines with useless calls demonstrated the ability of malicious code to reach out to other key infrastructures and cause serious damage. Another potential content-based threat that may soon enter the wireless world, as wireless devices become more sophisticated over time, is the embedded script virus. Prior to the first observation of this class of viruses, viruses could be contracted only through e-mail by double clicking on an infected e-mail attachment. With the discovery of embedded script viruses,